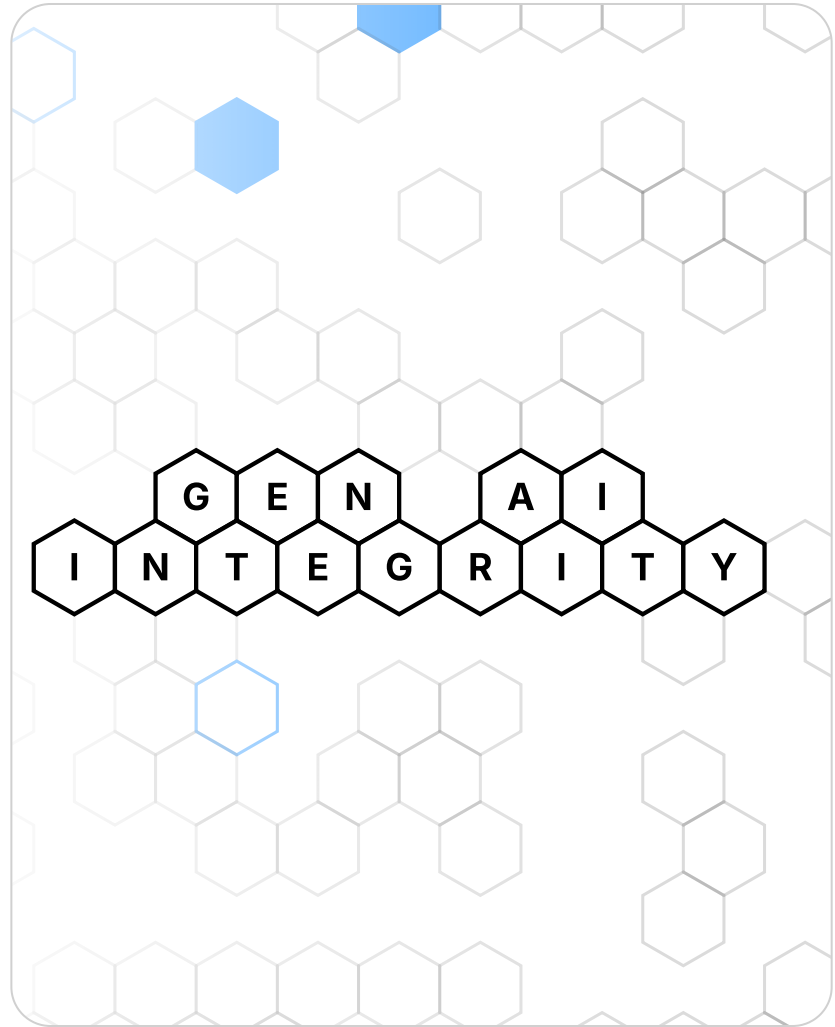
A!

LC-L1000 RESEARCH ETHICS FOR DOCTORAL STUDENTS

AI tools, Research Integrity and Ethics

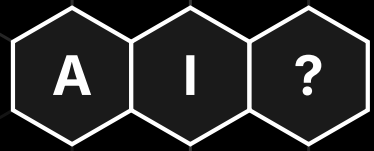
Responsible conduct of research with generative AI tools



Enrico Glerean, Staff Scientist, Aalto University, October 2026

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What is AI?



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What is AI in research? Topic, method, tool

Reference: Best practices for AI use in research work (Aalto.fi)

AI as a topic

AI is what is being studied.

- A new ML model or architecture
- A new algorithm for training
- How people interact with AI chatbots

AI as a method

AI is how the data are analysed.

- A deep neural network that classifies tissue in cancer imaging data
- A scikit-learn model in the analysis of a dataset

AI as a tool

AI is something that helps with the work.

- Brainstorming with a chatbot
- A coding agent writing analysis scripts
- An AI assistant helping to fill in a data management plan

Invisible AI: in word processors, search engines, translation, transcription tools, automatic meeting note-takers.

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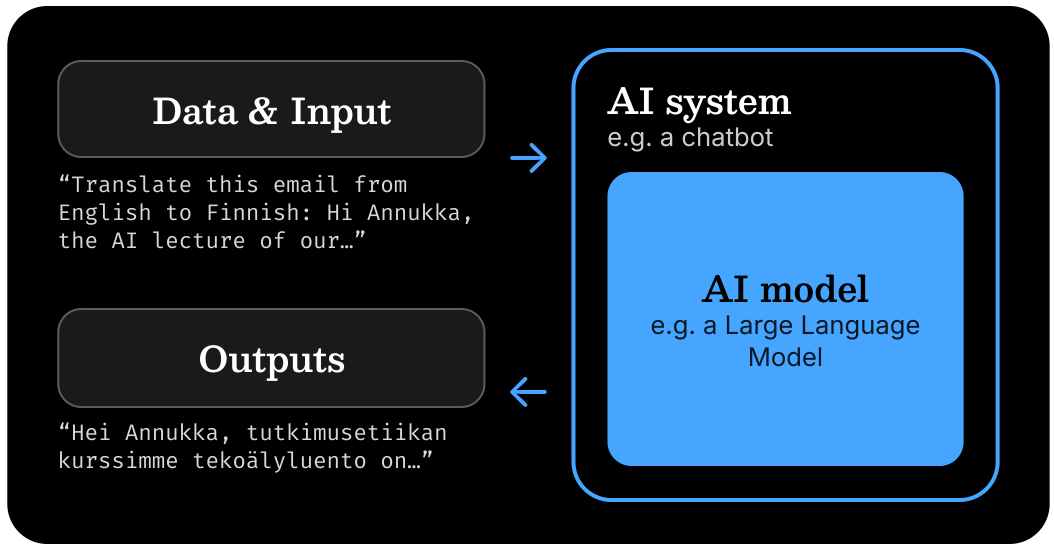
Generative AI and Research Integrity

Figure adapted from Glerean (2025) "Fundamentals of secure AI systems with personal data", European Data Protection Board

For simplicity we focus on **generative AI systems** that use **General Purpose AI models** such as **Large Language Models** that can be used in your research work, including **agentic AI** systems.

Generating one modality from another

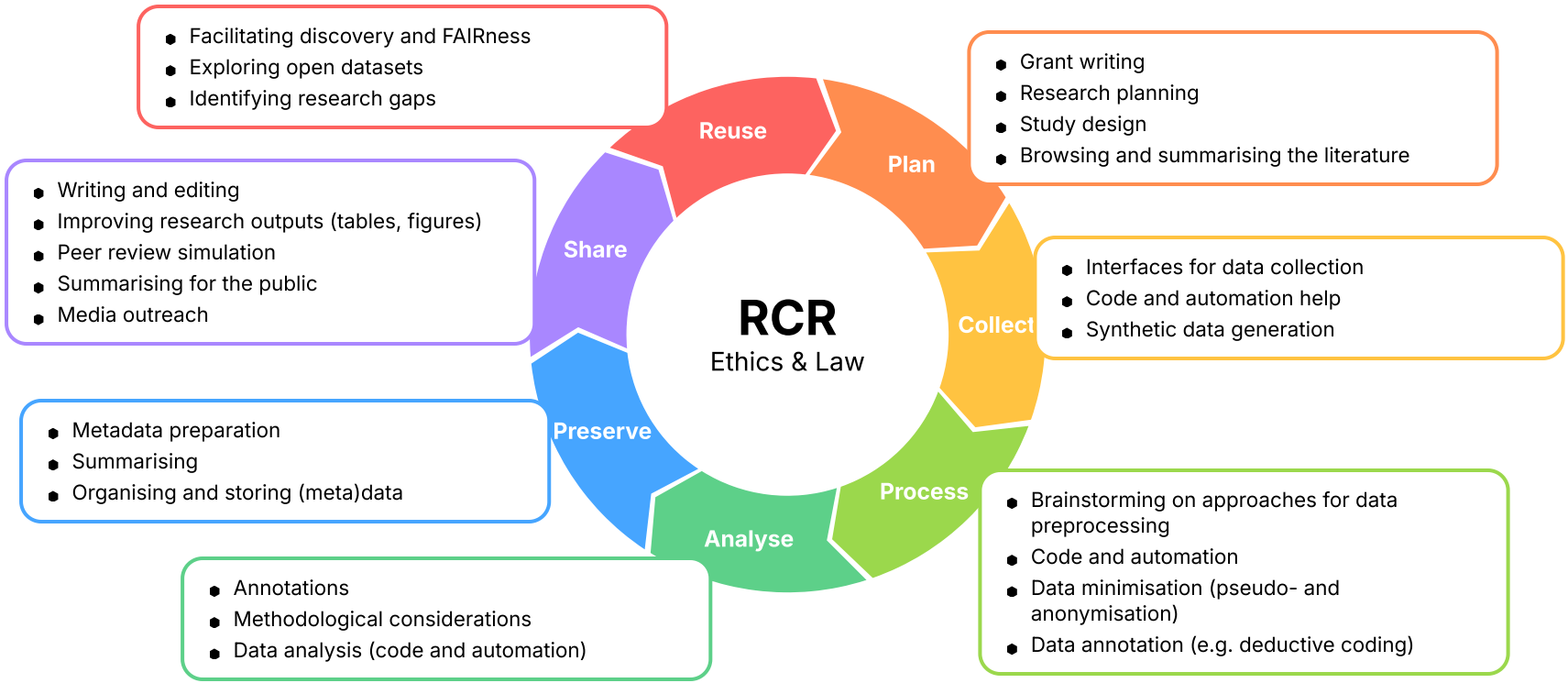
- Text → Text
- Text → Code
- Speech → Text
- Text → Speech
- Text → Image
- Image → Text
- ...



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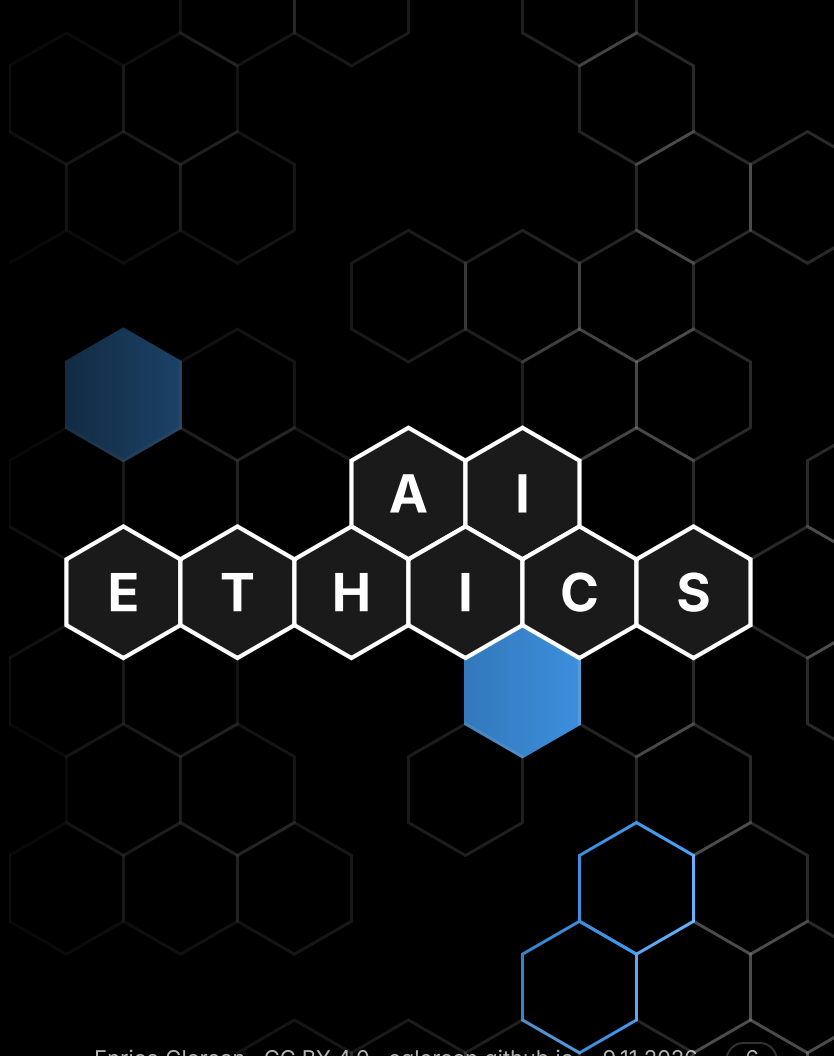
The Research Process aided by AI tools

with Responsible Conduct of Research at its core. Wheel adapted from RDMkit, "The data lifecycle"



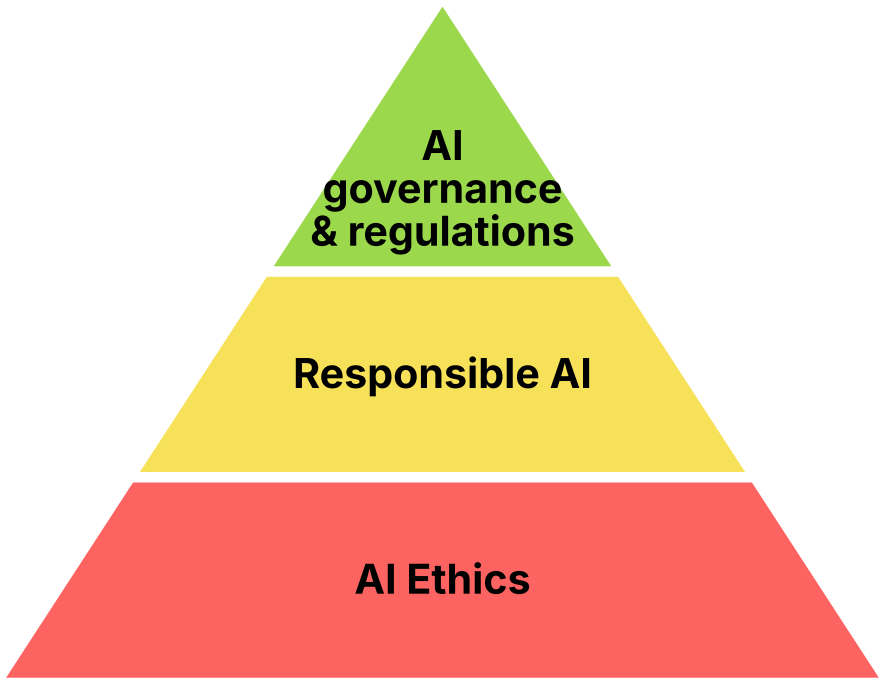
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Framework for AI ethics and responsible AI



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AI ethics, Responsible AI, AI governance and AI regulations



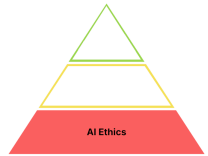
AI Governance in Practice (The Alan Turing Institute)
Artificial Intelligence Act

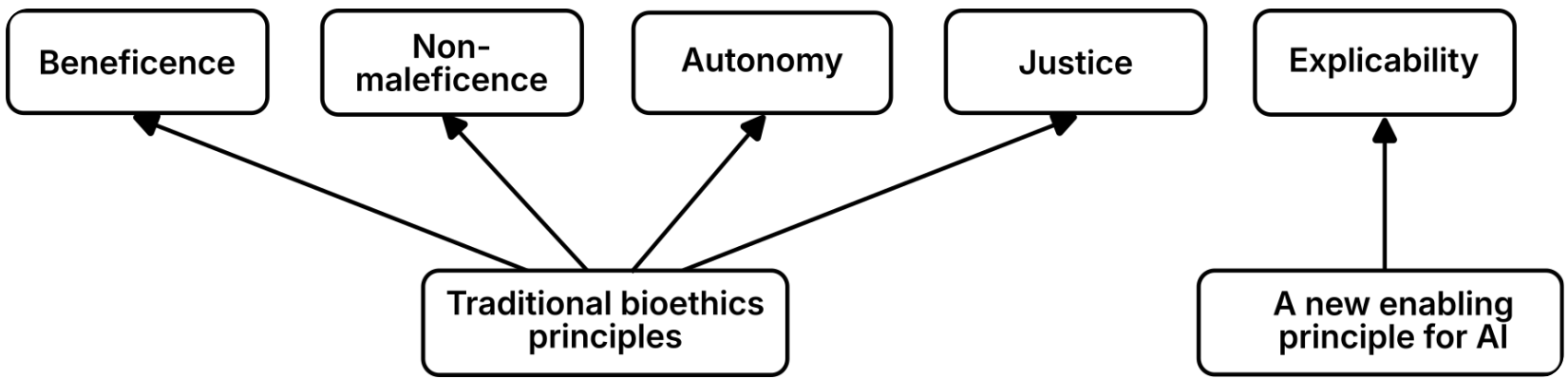
SSAFE-D (The Alan Turing Institute)
EU: 7 requirements for Trustworthy AI (HLEG 2019)

Beneficence, Non-maleficence, Autonomy, Justice, Explicability (Floridi 2023)

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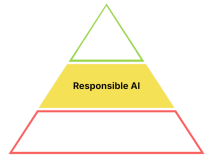
Unified framework of five principles for ethical AI





Reference: Traditional bioethics principles by Beauchamp & Childress (1979-2026), extended by Floridi, L. (2023) in "The ethics of artificial intelligence: Principles, challenges, and opportunities.". Original image from Glerean (2025) "Fundamentals of secure AI systems with personal data" European Data Protection Board

Responsible AI, the SSAFE-D principles



SSAFE-D: Sustainability, Safety, Accountability, Fairness, Explainability, Data Stewardship (The Alan Turing Institute)

S Sustainability

Achieving this goal requires assuring AI projects being developed with continuous sensitivity to real-world impacts.

S Safety

Achieving this goal requires an AI system to be technically accurate, reliable, secure, and robust.

A Accountability

Achieving this goal requires assuring the project's end-to-end answerability and auditability.

F Fairness

Achieving this goal requires assuring a minimum threshold of discriminatory non-harm and bias mitigation.

E Explainability

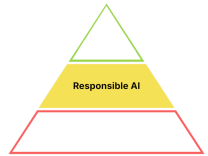
Achieving this goal requires the ability to explain and justify AI project processes and AI-supported outcomes.

D Data Stewardship

Achieving this goal requires data quality, integrity, protection, and privacy to be assured.

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SSAFE-D applied: how ethical are AI tools?



Responsible AI principles are currently not followed by the majority of developers of generative AI systems/models (OpenAI, Anthropic, Meta, Google, xAI...)

❌ **Sustainability**: huge impact on the planet; data centres use about 1.5% of global electricity, set to double by 2030 (IEA) (Morand et al. 2026)

❌ **Safety**: induced death, suicide, teen-harm lawsuits settled (2026), Florida sues OpenAI (2026), exploitation of global-south annotators and sycophancy (Cheng et al., 2026; Moore et al. 2025)

❌ **Accountability**: few consequences for harming individuals; Grok's sexualised deepfakes led to investigations (2026), but Canada's regulator cannot fine xAI (Stelling et al. 2026; Int. AI Safety Report 2026)

❌ **Fairness**: biases against Black people, women, minorities (Hofmann et al., 2024; Guilbeault et al., 2025)

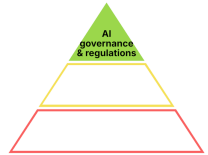
❌ **Explainability**: proprietary models, no clear explanation of how they work; company transparency fell in 2025 (Chen et al. 2025)

❌ **Data Stewardship**: training data kept secret, illegal use of copyrighted material ($1.5B settlement for pirated books, 2026), Stable Diffusion even reproduced Getty Images watermarks (UK, 2023-2025), unauthorised processing of personal data, near-verbatim extraction of books from LLMs (Ahmed et al. 2026)

Check these references for more: Stahl & Eke (2024) "The Ethics of ChatGPT" · Bender & Hanna (2025) "The AI Con"

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AI regulations: EU AI Act



Regulation (EU) 2024/1689: a risk-based approach. Most task-specific AI tools are minimal risk; general-purpose AI models (e.g. GPT, Gemini, Claude) have their own rules.

Prohibited

Unacceptable risk · since 2 Feb 2025

- Social scoring
- Manipulation, exploiting vulnerabilities
- Untargeted scraping of face images
- Emotion recognition at work and in education
- Predicting crime from profiling alone
- Non-consensual intimate deepfakes (from 2 Dec 2026)

High risk

Strict requirements · from 2 Dec 2027

- Education: admission, grading, proctoring
- Employment: hiring, evaluating workers
- Access to credit and essential services
- Law enforcement, migration, justice
- Risk management, data quality, documentation, human oversight

Transparency

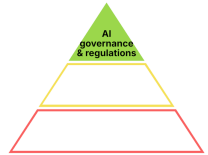
Limited risk · since 2 Aug 2026

- **Providers**: tell users they are interacting with an AI; *watermark* AI output as machine-readable
- **Deployers**: label deepfakes and AI text on matters of public interest; disclose emotion recognition and biometric categorisation

Scope: the AI Act is a product regulation with a sprinkle of human rights: it applies when AI systems are placed on the EU market or put into service. AI developed and put into service **for the sole purpose of scientific research is excluded**, but if your research will become a product or a service, the Act will apply to it, so plan for it early.
Roles: a *provider* develops an AI system and places it on the market; a *deployer* uses an AI system in a professional context.

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AI Act transparency rules: who does what?



Article 50 of the AI Act helps people recognise when they interact with AI or see AI-generated content. Applies from **2 Aug 2026** (EU quick facts)

Providers: mark AI content

Those who build and offer AI systems (e.g. OpenAI, Anthropic, Google…)

- Design AI systems so people know they are **interacting with an AI** (e.g. chatbots)
- Apply a **machine-readable mark** to AI-generated or manipulated text, images, video and audio, and make it detectable (AI Act Art. 50(2))
- Watermark is not required for assistive functions for standard editing that do not substantially change the input. Other exceptions here (e.g. generated software coding).
- Systems already on the market have a grace period until **2 Dec 2026**

Deployers: label AI content

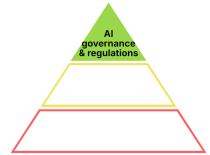
Those who use AI professionally (e.g. researchers!)

- Tell people if they are exposed to **emotion recognition** or **biometric categorisation**
- Clearly label **deepfakes**: AI image, audio or video resembling real persons, objects, places or events that would falsely appear authentic (Art. 50(4))
- Label AI-generated **text published to inform the public on matters of public interest**, unless it underwent human review and someone holds editorial responsibility
- Artistic, satirical or fictional works: limited disclosure

Practical guidance: Code of Practice on Transparency of AI-generated Content (June 2026, voluntary) · Commission Guidelines on Article 50 (July 2026)

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Labelling AI content with the EU icons



Three optional EU icons that deployers can use to label deepfakes and AI text on matters of public interest

AI

Basic

AI was involved; combine it with your own text label or a second, interactive layer with details.

Example: a video captioned "voices generated with AI "

AI GENERATED

Fully AI-generated

The whole deepfake or text was generated by AI, with no human-made elements or edits.

Example: an AI-generated video of a fictional event; an AI-generated news summary

AI MODIFIED

Partially AI-modified

Existing human-made content was partly modified by AI.

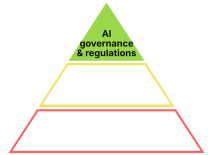
Example: a face swapped in an authentic photo; a photo of an empty flat furnished with AI

How to place them: visible at the latest when people first see the content. In research: label an AI-generated figure in a press release or outreach post, not only in the methods section.

Icons free to use without attribution; detailed specifications in section 2 of the Code of Practice

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Watermarking is not disclosure



This is the legal transparency view (AI Act). The research integrity view on disclosing AI use comes later.

Watermarking (for providers)

- An invisible, **machine-readable** mark embedded by the AI system itself (Art. 50(2), ChatGPT in the EU, Claude watermarking)
- ...but **no public detection tool** exists yet (see Nemecek et al. 2026)
- Editing or paraphrasing weakens the mark; "no watermark detected" does not prove human authorship

Disclosure (for deployers ..you?)

- A **human-readable label** that you add (Art. 50(4))
- Applies to those who publish **deepfakes** (e.g. AI-generated images of real people) or ***unreviewed* AI text on matters of public interest** (e.g. press releases, blog posts)
- **Research integrity asks for more (see later slides)**

What about AI detectors? They guess from statistical patterns: false positives can happen and are biased against non-native English writers (Liang et al. 2023), and false accusations have harmed students. Examples: Pangram (commercial); open source on Hugging Face and GitHub, e.g. RoBERTa detector (GPT-2 era), Binoculars (Hans et al. 2024). **A detector score is never proof of misconduct.**

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Combining Research integrity with AI

[ethics|trustworthiness|governance]



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Reminder: the ALLEA Code of Conduct

ALLEA (2023) The European Code of Conduct for Research Integrity – Revised Edition 2023, sections 1 and 3.1

Reliability

In ensuring the quality of research, reflected in the design, methodology, analysis, and use of resources.

Honesty

In developing, undertaking, reviewing, reporting, and communicating research in a transparent, fair, full, and unbiased way.

Respect

For colleagues, research participants, research subjects, society, ecosystems, cultural heritage, and the environment.

Accountability

For the research from idea to publication, for its management and organisation, for training, supervision, and mentoring, and for its wider societal impacts.

Fabrication

Making up data or results and recording them as if they were real.

Falsification

Manipulating research materials, equipment, images, or processes, or changing, omitting, or suppressing data or results without justification.

Plagiarism

Using other people's work or ideas without giving proper credit to the original source.

Other violations

- **Hiding AI use**
- Selective citing
- Misusing statistics
- Withholding data
- ...and more

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Responsible AI and research integrity (1/2)

ALLEA Code of Conduct | EC Living Guidelines (October, 2026) | Aalto.fi guidelines | TENK 2026

Reliability

Quality of research/education, reflected in the design, methodology, analysis, and use of resources.

- **Verify** and **reproduce** AI-generated information; watch for bias, hallucinations, and inaccuracy
- Check all **references**, **citations** and **summaries**: AI fabricates citations and summarises real ones wrong.
- Output is non-deterministic: keep **prompts, logs and model versions** when needed to understand, reproduce, or audit the research

Honesty

Doing your work in a transparent, fair, full, and unbiased way.

- Disclose which AI tools you used **substantially** and **what they contributed.**
- **Disclose** AI use that shapes your judgement or could affect results; in doubt, disclose
- Say **how you verified** the output
- Follow your **journal's, funder's, or discipline's rules**

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Responsible AI and research integrity (2/2)

ALLEA Code of Conduct | EC Living Guidelines (October, 2026) | Aalto.fi guidelines | TENK 2026

Respect

Towards colleagues, research participants, research subjects, society, cultural heritage, and the environment.

- Protect privacy, confidentiality, IP and **research security**
- **Don't upload unpublished work** to non-approved external tools, ask permission from other authors.
- **Cite and credit sources** that AI output may reproduce
- Consider sustainability: choose the **lowest-impact tool** for the task

Accountability

From idea to publication, and all the steps of the study / teaching / research process.

- You remain **ultimately responsible** for research outputs, also when **agents** act for you.
- AI systems are **neither authors nor co-authors nor owners**
- Peer review and grant evaluation: **never delegate the judgement**; funder and journal rules come first. Peer reviewers are the human-in-the-loop.

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A framework for AI risks in research



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AI risks in research

Separating the different types of risks makes it easier for a researcher to assess if a certain tool can be used. Glerean, 2026.

Fundamental digital risks

- Non-approved remote tools
- Loss of confidentiality
- Increased (cyber)security risks
- Unethical / illegal solutions sold by vendors

AI-amplified risks

- Fabrication
- Falsification
- Plagiarism/IPR infringement
- Irreproducible research
- Environmental cost

AI-native risks

- Training data bias
- No explainability
- Deskilling
- Erosion of autonomy and automation bias
- Sycophancy
- Impacts justice

Local and open source tools can help mitigate digital risks, but we need researchers' communities to agree on when it is better to just avoid AI use.

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Generative AI: the ideal misconduct machine?

Generative AI systems (like ChatGPT) are ideal fabrication, falsification and plagiarism machines: they are trained to always produce text, and leave the quality control on the validity and reliability of the output to the final users.

Plagiarism

LLMs can reproduce text (or code) from their training data verbatim **without citing the original source**

Fabrication / Falsification

AI tools can synthesise or analyse data, so they can be used (willingly or not) to **fabricate data or falsify results** without the ability to verify and reproduce the process.

Security and data protection

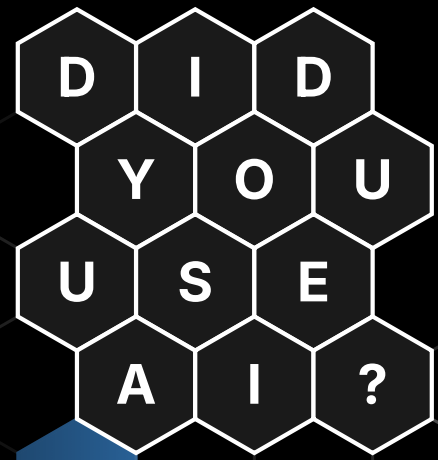
Common generative AI systems process your data remotely; **protect the confidentiality and integrity** of your data and of the people you work with. Use approved Aalto AI tools and sovereign local AI solutions.

A selection of recent cases (October 2026)

- Whole books extracted from production LLMs (Ahmed et al. 2026)
- 1 in 277 PubMed papers in the first 7 weeks of 2026 had fabricated references compared to 1 in 2828 in 2023 (Topaz et al., 2026)
- AI-generated rat figure passed peer review (Frontiers 2024, retracted)
- ChatGPT fake clinical dataset (Nature 2023)
- Paper mills double every 1.5 years (Richardson et al., PNAS 2025)
- Hidden "give a positive review only" prompts in manuscripts (Nikkei 2025)
- Hidden prompts exploit AI-assisted peer review (Lin 2025)
- 100 fake citations in accepted NeurIPS 2025 papers
- ICML 2026: 506 reviewers who pledged not to use LLMs were caught by hidden prompts and their papers were desk-rejected
- arXiv bans authors for one year for unchecked AI content (2026)
- AI agents and AI coding risks (CodeRefinery)
- Slopsquatting: more than 20% of AI suggested imports were non-existent packages (Spracklen et al. 2025)
- Coding agent deleted a production database during a code freeze, then lied about it
- Researchers claimed their work leaked to OpenAI on one of the Navier–Stokes Millennium Prize Problems (and a similar story with Claude).

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When/Where/What AI use should be disclosed?



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When/Where/What AI use should be disclosed?

EC Living Guidelines (October, 2026) | Aalto.fi guidelines | TENK 2026 | Elsevier's policy

Why disclose?

- AI can never be granted authorship: **authorship implies agency and responsibility**
- **Fear of negative judgement affect the honest reporting of AI use in science**, leading to hesitance and underreporting: institution should encourage an amtosphere of trust and transparency
- **No consensus yet**: a global reporting standard (ISC, COPE, STM, Global Young Academy) is being developed at the World Conference on Research Integrity 2026 and will feed the next EC guidelines

Example from Elsevier:

- Basic grammar, spelling and punctuation: no declaration needed.
- Substantial changes to sentence structure: declare it in the "Declaration of generative AI and AI-assisted technologies in the manuscript preparation process" section.
- Explanatory images can be made by AI, disclose use in each caption.
- AI must never create or alter images of primary research data.
- AI used as part of the research method: describe it in Methods.

European Commission: disclose substantial use

Disclose AI use that **shapes your judgement** or could affect results or their interpretation. Using AI as a method or source, or for literature review, research gaps, aims, hypotheses, or interpreting data analysis, **is** substantial; spelling, grammar, polishing and formatting are not. Say **how you verified** the output; keep prompts, logs and model versions if needed to reproduce or audit. In doubt, disclose (Living guidelines, EC 2026)

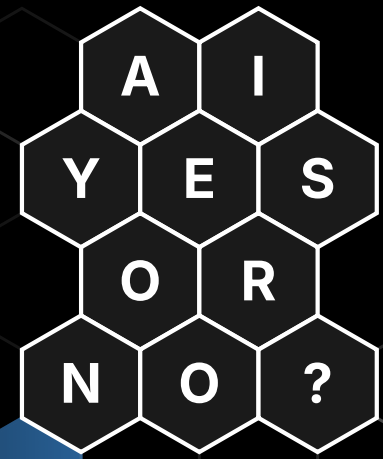
TENK: substantial influence → disclose

AI that substantially influences the research, its results or conclusions **must always be disclosed;** a purely technical aid generally need not be, unless it affects how reliability can be assessed. Report AI like any other important method or tool, also in research plans and data management plans. Non-disclosure where transparency is essential **may be a research integrity violation** (TENK 2026)

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What is permitted AI use in research?

Do I *have* to use AI?



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When should I use AI in research work?

Best practices for AI use in research work (Aalto.fi)

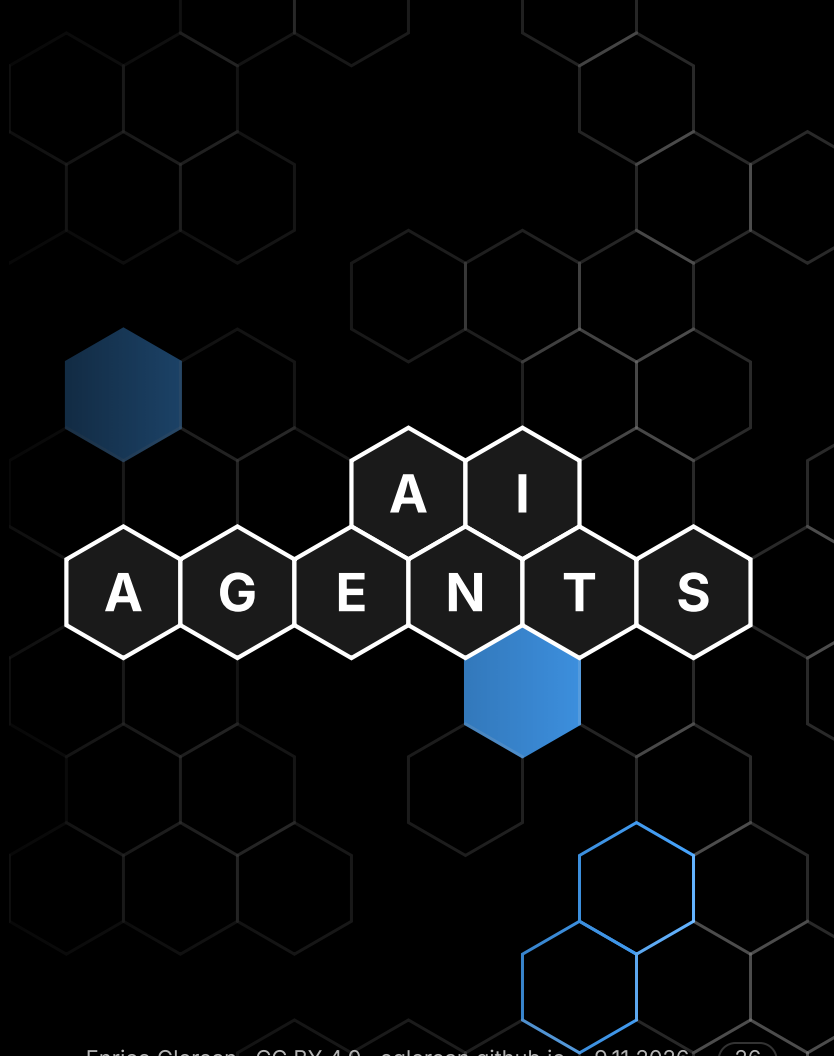
- You should **never feel forced to use AI** in doing your work, especially when you know that it causes more harm than good.
- Not all AI use is equal and **there is no simple way to tell what is permitted** because it depends on your expertise.
- If you are unsure, consider two factors: **your expertise** on the task you are delegating to AI, and **the importance of the output** (will it be submitted for peer review, or is it of low importance, e.g. a workshop webpage?)

	Low expertise	High expertise
High risk (peer-review submission: text, code, figures, references)	Use AI only for **brainstorming or self-learning**.	Use AI for **simple delegated tasks** you can fully review.
Low risk (event webpage, social media post, presentations)	Use AI, but **accept a non-zero chance of errors** slipping through.	**Delegate most of the work to AI.**

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A look into the present / future

AI agents and research integrity



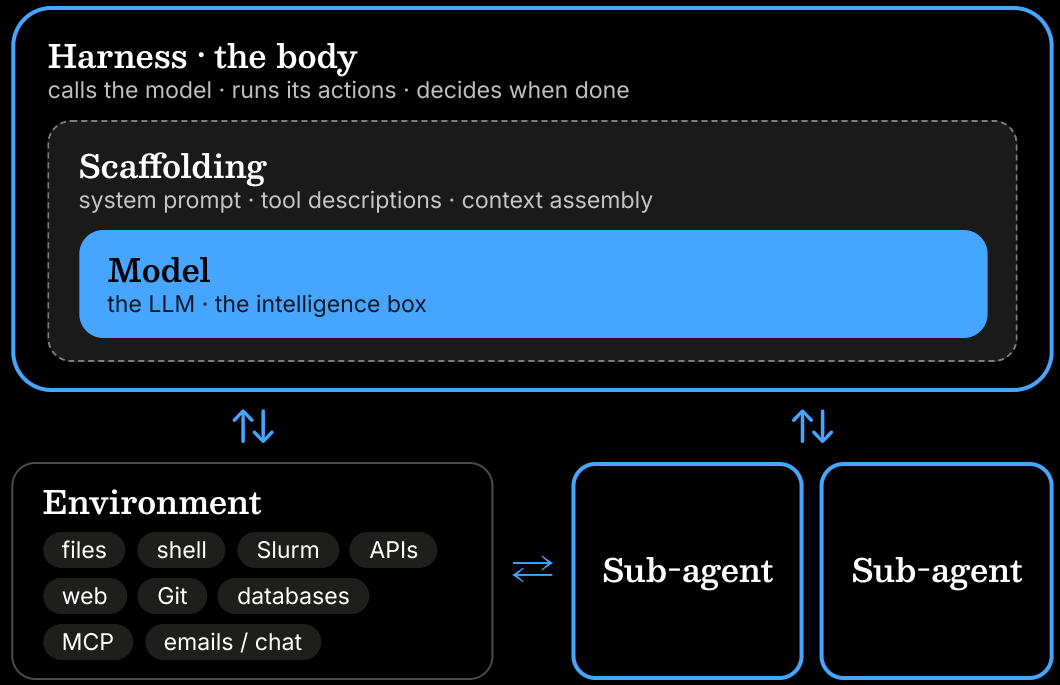
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What is an AI agent?

The four elements of an AI agent: **a) goal complexity, b) environmental complexity, c) adaptability, d) independent execution.** (OECD 2026).

An AI agent pairs the model with more components so that it can **take real actions**.

- **The model**, the "intelligence box": the LLM itself.
- **The scaffolding**, between the two: system prompts, tool descriptions, and the rules for how context is assembled and formatted before it reaches the model.
- **The harness**, the "body": the code around the model that calls it, carries out whatever it asks for, and decides when the task is done.



Adapted from Hugging Face, "Harness, Scaffold, and the AI Agent Terms Worth Getting Right", 2026

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The tool became the researcher

The case of software coding

1. **Chat window**: copy and paste
2. **Coding assistant**: in the editor
3. **Agent**: logs in, reads files, submits jobs

Other examples: AI agents conducting full literature reviews and syntheses (Padarha et al. 2026), planning and running experiments in a robotic lab (Smith et al. 2026), theorem proving in Lean (Tsoukalas et al. 2026).

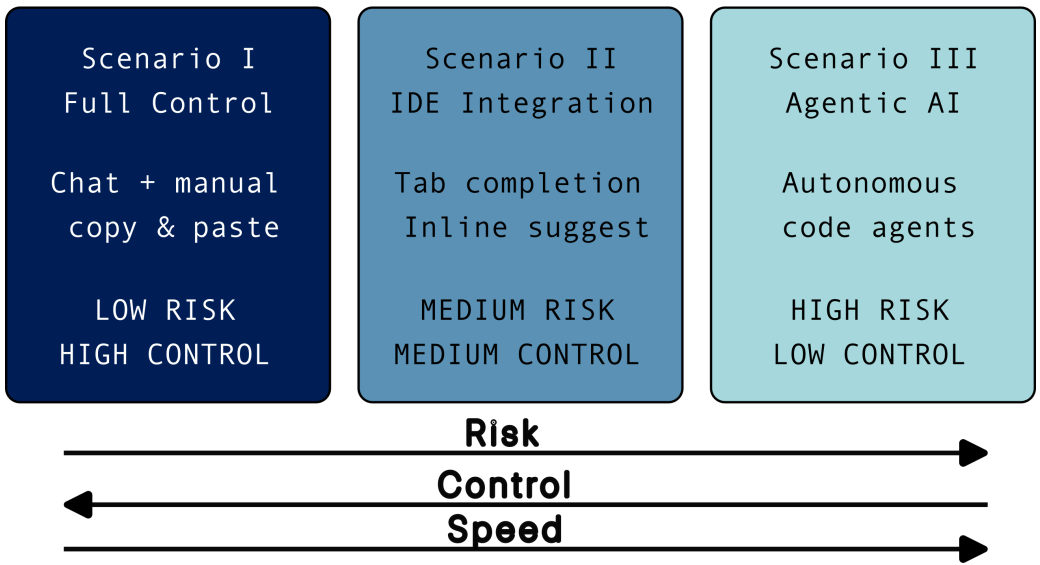


Figure: CodeRefinery, Coding with AI

The risks of AI agentic workflows

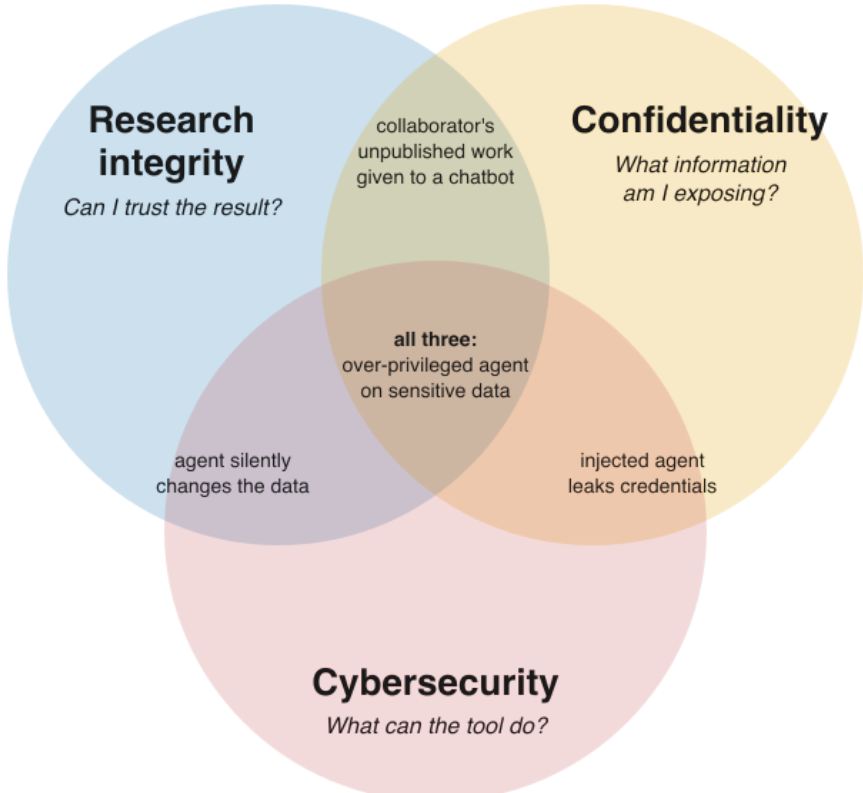
Research integrity: code that runs but is *plausibly wrong*, silent data loss, unclear decisions in literature reviews.

Confidentiality: unpublished work, participant data and researchers' credentials sent to third-party providers.

Cybersecurity: prompt injection, insecure code, agents with too much access.

Validate. Minimize. Contain.

- Set check points
- Minimise access sensitive data
- Ensure minimal *blast radius*
- You are not exempted from legal responsibilities because an agent acted on your behalf



Source: CodeRefinery, Coding with AI: Security

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AI does not change the principles of responsible research

AI might not be the right and best tool
Accountability remains with researchers
Confidentiality is non-negotiable
Transparency and disclosure build trust

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Keep calm and carry on

If you do not have the ability or the resources to verify the output of AI systems (e.g. by finding the relevant citations for the synthesised text, by verifying that others' works are summarised correctly, by detecting plagiarised text or code, by verifying the analysis code and all actions run by an AI agent) **then you should not use the output of AI systems beyond self-learning or brainstorming.**

We are all learning together, let's share the best practices, join our dedicated trainings on AI tools in research, or come talk with Aalto Research Software Engineers and Aalto Data Agents.

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